

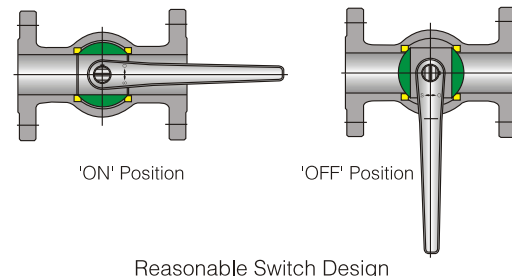
Floating Ball Valve

Structural Features

Float ball valves are superior products developed upon the advanced technology home and abroad. Provided with updated design ideas, they meet the requirements on safety and convenience in general industrial standards.

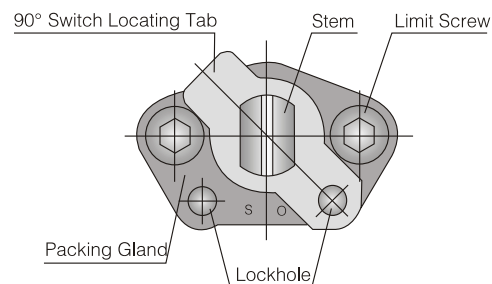
1. Switch Indication of Hand Operated Float Ball Valve

As the moving parts of valve, ball, valve stem and handle are an assembly unit. The head of valve stem is shaped a diploid, thus to easily distinguish whether valve is in the 'ON' position or 'OFF' position from handle position. When handle or stem diploid is in parallel with the pipe axis, valve is in 'ON' position; when handle or stem diploid is vertical to the pipe axis, valve is in 'OFF' position.



2. Valve Locking Device

To prevent misoperation, the fully opened or closed position of valve can be locked up, especially when valves are mounted outdoors or when valves are not allowed to be opened or closed by technical process. To prevent misoperation by other persons, it seems very important to have valve position locked up. Thereby, lockhole locating ab is designed if requested by users to meet the technological requirements of users.



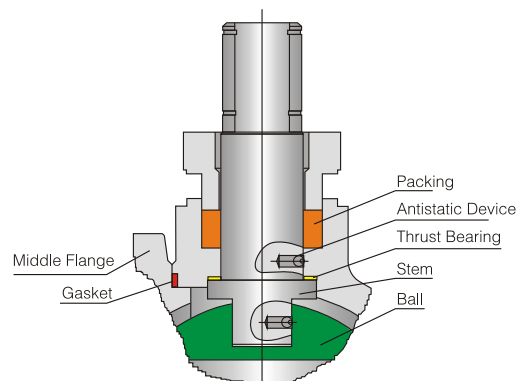
90° Switch with Lockhole Locating Tab

3. Anti-flyout Structure of Valve Stem

When medium passes through the valve, the pressure in valve body may possibly push the stem out, or when the valve is in repair, if there is pressure in the middle cavity, the stem or medium seems to easily fly out upon disassembling the valve, which can cause injuries to human beings. To prevent these possibilities, a dummy club is placed at the lower part of the stem. In this way, even if there is fire, packing and thrust bearing are burnt or packing is damaged due to other causes, the pressure of medium in valve body will make the dummy club of valve stem in close contact with the upper seal face of valve body, thus to prevent medium leaking out from the damaged packing position.

4. Antistatic Device

When operating the valve, the friction between the ball and the nonmetal seat, like PTFE or etc. will produce electrostatic charge that can be accumulated on the ball. To prevent static spark, an antistatic device is placed on the valve to derive the electric charge accumulated on the ball from the static channel between the ball and the stem, or between the stem and valve body.



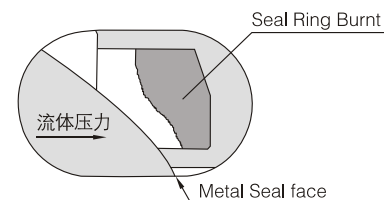
Stem Anti-flyout Structure
Stem Antistatic Device
Middle Flange Leak-tight Structure

5. Fire Protection Structure

In case of fire, the nonmetal packing or seat (non-fireproof material) will be burnt, the considerable medium leakage may possibly cause fire spread. Here, the fire protection structure functions to prevent medium from mass leakage. As shown in the figure, once the seat is burnt, the ball will directly contact the metal face on valve body, thus to prevent medium leaking out from the burnt seat. The design of fire protection and antistatic structure shall conform to the requirements of API607, JB/T6899 and the second part of BS6755.

6. Middle Flange Leak-tight Structure

The connection between valve body and the left body is sealed by gasket. To prevent seal leakage resulted from fire, high temperature or vibration etc., the valve body and the left body is designed metal-to-metal contacted, thus to form up a seal flange to ensure leak-tightness.



Fire Protection Structure of Float Ball Valve

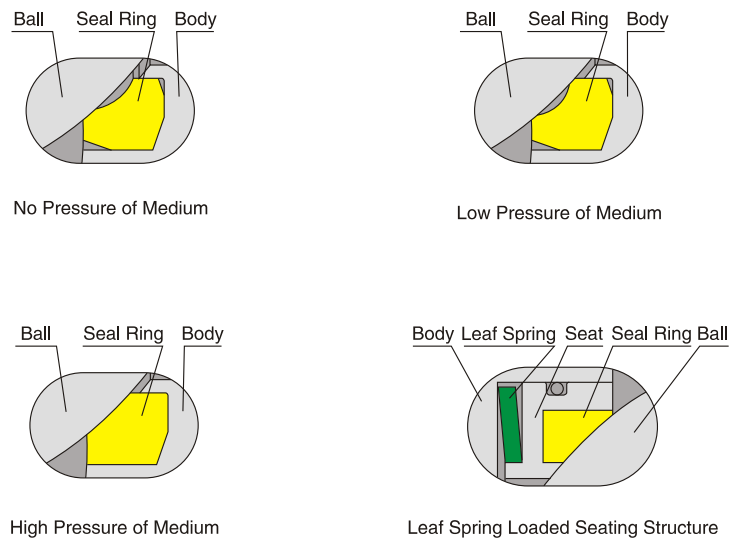
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7. Dependable Seating Structure

The two-way sealing structure embodying the years of our manufacturing experience in ball valves and the internationally advanced technology can release the pressure at valve seat and ensure reliable seal under high or low pressure and vacuum state. When the pressure of medium is low, the contact area between valve seat insert and ball is relatively small, thus providing high sealing load to ensure dependable seating seal. When the pressure of medium is high, the contact area between valve seat insert and ball is relatively large, so that the valve seat insert can endure the considerable medium thrust and cannot be damaged.

Regarding ball valves of very low working pressure, in consideration that the pressure of medium cannot guarantee the dependable seal of valve seat, and that the pretightening force will decrease after a long time of service, so we apply leaf spring loaded seating structure for valves working under low and ultralow pressure or under vacuum conditions, thus to ensure persistent and dependable seal.

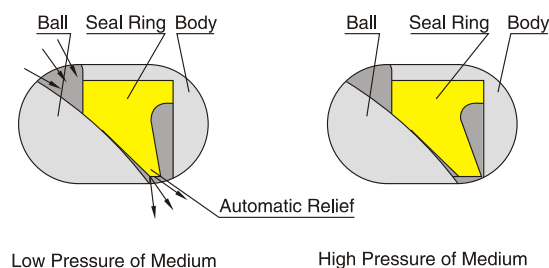
The seating structure of medium and high temperature ball valves use PPL for seal ring that can serve a temperature of 300°C. Ball valves with metal seal and high temperature resisting materials can be used to deal with even higher temperature. This structure takes the function of one-way seal.



Double Bevel Elastic Seal Ring of Float Ball Valve

8. Self-relieving Structure

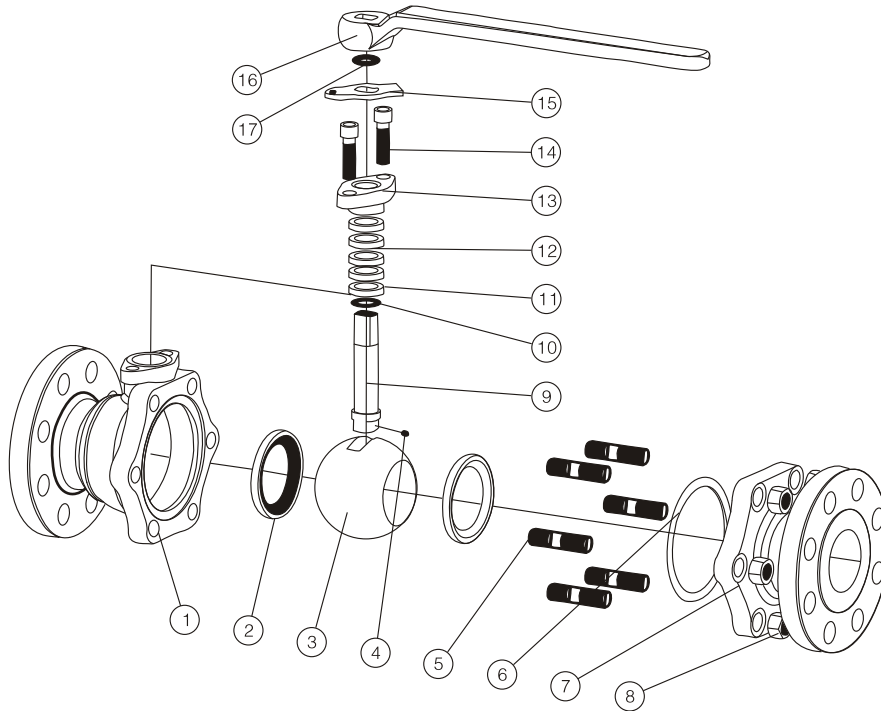
In case of abnormal rise of pressure in the middle cavity of valve resulted from the medium left there suffered from changes of pipeline pressure or medium temperature, automatic pressure relief of valve seat can be effected under the pressure of the medium itself, thus to guarantee the safety of valve.



V-slot Elastic Seal Ring of Float Ball Valve

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Floating Ball Valve Structural Drawing

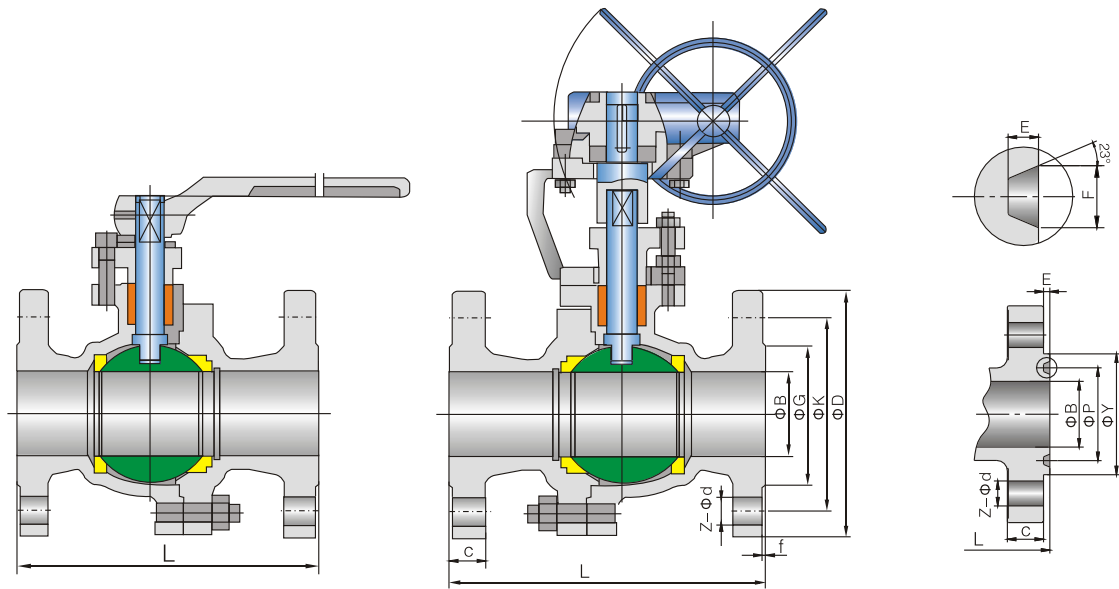


The Material of Main Parts

NO.	Parts Name	Carbon Steel Series	Stainless steel Series	Low Temperature Series	Anti-Sulphur	
					Carbon Steel	SS Series
1	Body	A216 WCB	A351、CF8、CF8M、CF3、CF3M	A352 LCB、LCC	GB/T12229、A216 WCB	A351 CF8M
2	Seat	PTFE,RPTFE,Braided Graphite Fibre ,Steel+Rubber Assemble				
3	Ball	A105+HCr/ENP	A351、CF8、CF8M、CF3、CF3M	A352 LCB、LCC+ENP	A105+HCr/ENP	A351 CF8M+ENP
4	Spring	INCONEL 750				
5	Stud	A193 B7	A193 B8、B8M	A320 L7	A193 B7M	A193 B8M
6	Gasket	Soft Graphite+SS				
7	Eyeblt Nut	A194 2H	A194 8M	A194 4	A194 2HM	A194 8M
8	Bonney	A216 WCB	A351、CF8、CF8M、CF3、CF3M	A352 LCB、LCC	GB/T12229、A216 WCB	A351 CF8M
9	Stem	A182 F6a	A182 F304、316	A182 F6a	A182 F304	A182 F316
10	Bush Bearing	Metal+PTFE; Braided Graphite Fibre				
11	Packing Seat	A182 F6a		A182 F6a	A182 F6a	
12	Packing	Soft Graphite,PTFE				
13	Nipple	A216 WCB	A351 CF8、CF8M	A351 CF8	GB/T12229、A216 WCB	A351 CF8M
14	Gland Packing	A193 B7	A193 B8、B8M	A320 L7	A193 B7M	A193 B8M
15	Locating Plate	ASTM A570+Zn(Cr)				
16	Circlip	AISI 1066				
17	Hand Lever	A216 WCB				

Note: The chart above only lists out some common composition of valve parts, We may provide other different parts material composition according to the customer's request or the actual valve working condition.

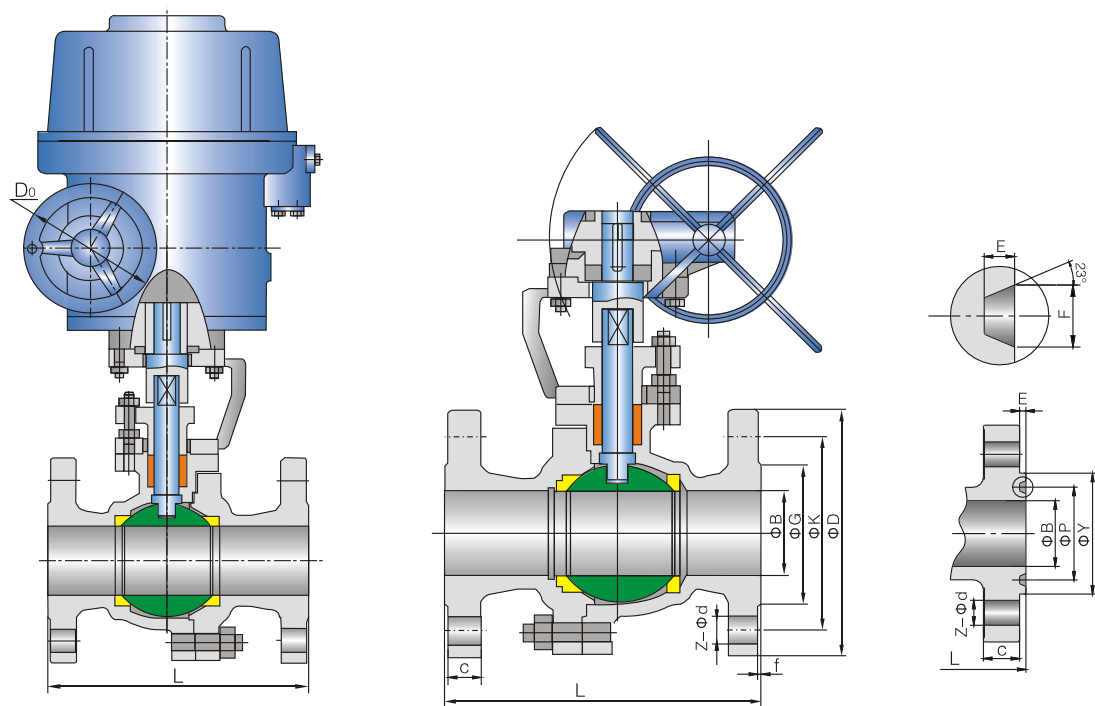
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Main Connection Dimension

Class	DN		L		B	D	G	f	K	z-Φd	C	RTJ			
	in	mm	RF	RTJ								Y	P	E	F
150Lb	1	25	127		25	108	50.8	1.6	79.3	4-15.9	11.2				
	1 1/4	32	140		32	117	63.5	1.6	88.9	4-15.9	12.7				
	1 1/2	40	165		38	127	73.2	1.6	98.6	4-15.9	14.2				
	2	50	178		49	152	91.9	1.6	120.7	4-19.1	15.8				
	2 1/2	65	191		62	178	104.6	1.6	139.7	4-19.1	17.5				
	3	80	203		74	191	127	1.6	152.4	4-19.1	19.1				
	4	100	229		100	229	157.2	1.6	190.5	8-19.1	23.9				
	6	150	394		150	279	215.9	1.6	241.3	8-22.2	25.4				
	8	200	457		201	343	269.7	1.6	298.5	8-22.2	28.4				
	10	250	533		252	406	323.9	1.6	362	12-25.4	30.2				
300Lb	1	25	165	178	25	124	50.8	1.6	88.9	4-19.1	17.5	70	50.8	6.35	8.74
	1 1/4	32	178	191	32	133	63.5	1.6	98.6	4-19.1	19.1	79.5	60.32	6.35	8.74
	1 1/2	40	191	203	38	155	73.2	1.6	114.3	4-22.2	20.6	90.5	68.27	6.35	8.74
	2	50	216	232	49	165	91.9	1.6	127	8-19.1	22.4	108	82.55	7.92	11.91
	2 1/2	65	241	257	62	191	104.6	1.6	149.2	8-22.2	25.4	127	101.6	7.92	11.91
	3	80	283	298	74	210	127	1.6	168.3	8-22.2	28.4	146	123.82	7.92	11.91
	4	100	305	321	100	254	157.2	1.6	200.2	8-22.2	31.8	175	149.22	7.92	11.91
	6	150	403	419	150	318	215.9	1.6	269.8	12-22.2	36.6	241	211.12	7.92	11.91
	8	200	502	518	201	381	269.7	1.6	330.2	12-25.4	41.1	302	269.88	7.92	11.91

Floating Ball Valve



Main Connection Dimension

Class	DN		L		B	D	G	f	K	z- Φ d	C	RTJ			
	in	mm	RF	RTJ								Y	P	E	F
600Lb	1	25	216	216	25	124	50.8	6.4	88.9	4-19.1	17.5	70	50.8	6.35	8.74
	1 1/4	32	229	229	32	133	63.5	6.4	98.6	4-19.1	20.6	79.5	60.32	6.35	8.74
	1 1/2	40	241	241	38	155	73.2	6.4	114.3	4-22.2	22.4	90.5	68.27	6.35	8.74
	2	50	292	295	49	165	91.9	6.4	127	8-19.1	25.4	108	82.55	7.92	11.91
	2 1/2	65	330	333	62	191	104.6	6.4	149.2	8-22.2	28.4	127	101.6	7.92	11.91
	3	80	356	359	74	210	127	6.4	168.3	8-22.2	31.8	146	123.82	7.92	11.91
	4	100	432	435	100	273	157.2	6.4	215.9	8-25.4	38.1	175	149.22	7.92	11.91
900Lb	1	25	254	254	25	149	50.8	6.4	101.6	4-25.4	28.4	71.5	50.8	6.35	8.74
	1 1/4	32	279	279	32	159	63.5	6.4	111.2	4-25.4	28.4	81	60.32	6.35	8.74
	1 1/2	40	305	305	38	178	73.2	6.4	124	4-28.6	31.8	92	68.27	6.35	8.74
	2	50	368	371	49	216	91.9	6.4	165.1	8-25.4	38.1	124	95.25	7.92	11.91
1500Lb	1	25	254	254	25	149	50.8	6.4	101.6	4-25.4	28.4	71.5	50.8	6.35	8.74
	1 1/4	32	279	279	32	159	63.5	6.4	111.2	4-25.4	28.4	81	60.32	6.35	8.74
	1 1/2	40	305	305	38	178	73.2	6.4	124	4-28.6	31.8	92	68.27	6.35	8.74
	2	50	368	371	49	216	91.9	6.4	165.1	8-25.4	38.1	124	95.25	7.92	11.91